Week – 2

SORTING TECHNIQUES

N.Hema Harika

CH.SC.U4CSE24231

1.Bubble Sort

Code :

```
#include <stdio.h>
void BubbleSort(int arr[] ,int n){
 int i,j;
  for(i=0;i<n-1;i++){
    for(j=0;j<n-i-1;j++){
      if(arr[j]>arr[j+1]){
          int temp =arr[j];
           arr[j]=arr[j+1];
           arr[j+1]=temp;
      }}}}
 int main() {
  int n,arr[100];
  printf("No.of elements :");
  scanf("%d",&n);
  printf("Enter array elements \n");
  for(int i=0;i<n;i++){
      scanf("%d",&arr[i]);
      }
  printf("array elements before sorting\n");
 for(int i=0;i<n;i++){
      printf("%d\t",arr[i]);
      }
  printf("\n");
  BubbleSort(arr,n);
  printf(" array elements after sorting\n");
 for(int i=0;i<n;i++){
  printf("%d\t",arr[i]);
  }}
```

Output:

```
amma@amma28:~/Downloads$ gcc bubbleSort.c -o bubbleSort
amma@amma28:~/Downloads$ ./bubbleSort
No.of elements :4
Enter array elements
5 2 6 1
array elements before sorting
5       2       6       1
 array elements after sorting
1       2       5       6
```

2. Insertion Sort :

Code

```c
#include<stdio.h>
void insertionSort(int arr[],int n)
{
  int i,key,j;
  for(i=1;i<n;i++)
  { key = arr[i];
    j = i-1;
  while(j>=0 && arr[j]>key)
  {
  arr[j+1] = arr[j];
  j = j-1;
  }
  arr[j+1] = key;
  }
}
int main()
{
 int n,i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter array elements:\n");
```

```c
  for ( i = 0; i < n; i++)
  {
      scanf("%d", &arr[i]);
  }
  printf("Array elements before sorting: ");
  for (i = 0; i < n; i++)
  {
      printf("%d ", arr[i]);
  }
insertionSort(arr,n);
printf("\nArray elements after sorting : ");
for (i = 0; i < n; i++)
{
      printf("%d ", arr[i]);
}
  printf("\n");
  }
```

Output

```
148 amma@amma36:~/Documents$ gcc insertionsort.c -o insertionsort
amma@amma36:~/Documents$ ./insertionsort
Enter number of elements: 5
Enter array elements:
45
12
33
78
23
Array elements before sorting: 45 12 33 78 23
Array elements after sorting : 12 23 33 45 78
```

3. Selection Sort

Code :

```c
#include <stdio.h>
void selectionSort(int arr[] ,int n){
 int i,j;
 for(i=0;i<n;i++){
 int minindex = i;
  for(j=i+1;j<n-1;j++){
   if(arr[j]<arr[minindex]){
    minindex = j;}}
  if(minindex!=i) {
   int temp = arr[i];
    arr[i] = arr[minindex];
    arr[minindex]=temp;
    }}}
int main() {
  int n,arr[100];
  printf("No.of elements :");
  scanf("%d",&n);
  printf("Enter array elements \n");
  for(int i=0;i<n;i++){
      scanf("%d",&arr[i]);}
  printf("array elements before sorting\n");
 for(int i=0;i<n;i++){
      printf("%d\t",arr[i]);}
  printf("\n");
  selectionSort(arr,n);
  printf(" array elements after sorting\n");
 for(int i=0;i<n;i++){
  printf("%d\t",arr[i]);
  }}
```

Output:

```
amma@amma28:~/Downloads$ ./selectionSort
No. of elements: 5
Enter array elements:
2
3
4
5
1
Array before sorting:
2       3       4       5       1
Array after sorting:
1       2       3       4       5
```

4.Heap Sort

Code:

```c
#include<stdio.h>
void swap(int *a,int *b)
{
 int temp = *a;
 *a = *b;
 *b = temp;
}
void heapify(int arr[], int n, int i)
{
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
    if (left < n && arr[left] > arr[largest])
    {
        largest = left;
    }
    if (right < n && arr[right] > arr[largest])
    {
        largest = right;
    }
    if (largest != i)
    {
        swap(&arr[i], &arr[largest]);
        heapify(arr, n, largest);
    }
}
```

```c
void heapsort(int arr[],int n)
{
int i;
 for(int i = n/2-1;i>=0;i--)
 {
  heapify(arr,n,i);
 }
 for(i=n-1;i>0;i--)
 {
  swap(&arr[0],&arr[i]);
  heapify(arr,i,0);
 }
}
int main()
{
 int n,i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter array elements:\n");
    for ( i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Array elements before sorting: ");
```

```
    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
heapsort(arr,n);
printf("\nArray elements after sorting : ");
 for (i = 0; i < n; i++)
 {
        printf("%d ", arr[i]);
 }
 printf("\n");
 }
```

Output:

```
hema-harika@hema-harika-VirtualBox:~/Downloads$ gcc heapsort.c -o heapsort
hema-harika@hema-harika-VirtualBox:~/Downloads$ ./heapsort
Enter number of elements: 5
Enter array elements:
50
23
45
67
12
Array elements before sorting: 50 23 45 67 12
Array elements after sorting: 12 23 45 50 67
```

5.Bucket Sort

Code :

```c
#include <stdio.h>
#include <stdlib.h>
#define max 20
#define BUCKETS 10
void insertionsort(float arr[], int size){
    int i, key, j;
    for (i = 1; i < size; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key){
            arr[j + 1] = arr[j];
            j = j - 1;}
        arr[j + 1] = key; }
}
void BucketSort(float arr[], int n){
    float buckets[BUCKETS][max];
    int bucketsize[BUCKETS] = {0};
    int i;
    for (i = 0; i < n; i++){
        int index = (int)(arr[i] * BUCKETS);
        if (index == BUCKETS) index = BUCKETS - 1;
        buckets[index][bucketsize[index]++] = arr[i];  }
for (i = 0; i < BUCKETS; i++) {
        insertionsort(buckets[i], bucketsize[i]);
    }
```

```c
 int k = 0, j;
    for (i = 0; i < BUCKETS; i++){
        for (j = 0; j < bucketsize[i]; j++){
            arr[k++] = buckets[i][j]; }}
}
int main(){
    int n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    if (n <= 0)  {
        printf("Invalid number of elements.\n");
        return 1;}
float *arr = (float *)malloc(n * sizeof(float));
    if (arr == NULL)  {
        printf("Memory allocation failed!\n");
        return 1;
    }
 printf("Enter array elements:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%f", &arr[i]);}
 printf("Array elements before sorting: ");
    for (i = 0; i < n; i++){
        printf("%f ", arr[i]);
    }
```

```
 BucketSort(arr, n);
printf("\nArray elements after sorting: ");
    for (i = 0; i < n; i++){
        printf("%f ", arr[i]);
    }
    printf("\n");
    free(arr);
    return 0;
}
```

Output:

```
hema-harika@hema-harika-VirtualBox:~/Downloads$ gcc bucketsort.c -o bucketsort
hema-harika@hema-harika-VirtualBox:~/Downloads$ ./bucketsort
Enter number of elements: 5
Enter array elements:
0.45
0.56
0.12
0.08
0.23
Array elements before sorting: 0.450000 0.560000 0.120000 0.080000 0.230000
Array elements after sorting: 0.080000 0.120000 0.230000 0.450000 0.560000
```